

Universal Solid-State Kinetic Reactor

A Distributed Cellular Framework for Damping, Harvesting, and Self-Restoration

The MKRT-Elite v1.3 is a multi-disciplinary platform designed for autonomous kinetic mitigation. By integrating fractal gyroid geometries, conductive polymer lattices, and mineral-encapsulated "slick" surfaces, the system provides a zero-maintenance solution for maritime, aerospace, and civil infrastructure.

I. Core Architecture

The structural foundation utilizes a **Fractal Gyroid Lattice** printed in CNT-doped PVDF. This geometry is optimized for the highest possible surface-area-to-volume ratio, facilitating both kinetic damping and thermal exchange.

$$\sin(x)\cos(y) + \sin(y)\cos(z) + \sin(z)\cos(x) = 0$$

II. Cellular Autonomy

Moving away from centralized power, the system utilizes **Sub-Floor Solid-State Supercapacitors**. Each cell is self-powered via piezoelectric energy harvesting from ambient vibrations, maintaining a "Reflex Logic" response that triggers hardening in <1ms upon impact.

III. Material Science & Eco-Security

- **UV Barrier:** Encapsulated TiO₂ and ZnO nanoparticles within the TPU matrix prevent molecular embrittlement without environmental leaching.

- **Bio-Deterrent:** Hexagonal Boron Nitride (h-BN) molecularly doped skin creates a "Slick-Film" that prevents biological adhesion via low surface energy.
- **Self-Healing:** Controlled low-amperage current allows for thermal annealing of micro-fractures, extending the operational life to 50+ years.

IV. Strategic Intent

This technology is intended for open distribution to revolutionize global infrastructure safety. It is designed to be manufactured locally using 3D printing, reducing supply chain dependency and fostering sovereign engineering capability.